

Sets

Introduction

Sets are **primitive** objects when doing classical, pen-and-paper mathematics:

- no *definition*;
- only *rules* about how these objects work (unions, intersections, etc.).

That's all you need: do you ever look at $S \rightarrow T$ as $S \subseteq T$?

We've seen that, for Lean, the **primitive objects** are types; yet

- sometimes we still want to speak about *sets* as collections of elements
- we want then to play the usual games.

Definitions

+++ Every set lives in a **given** type, it is a set of elements (*terms*) in it:

```
variable (α : Type) (S : Set α)
```

expresses that α is a type and S is a set of elements/terms of the type α . On the other hand,

```
variable (S : Set)
```

does not mean "let S be a set": it means nothing and it is an error.

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+++ A set coincides with the test-function defining it.

Given a type α , a set S (of elements/terms of α) is a *function*

```
S : α → Prop
```

so $(\text{Set } \alpha) = (\alpha \rightarrow \text{Prop})$.

- This function is the "characteristic function" of the set S ;
- the $a \in S$ symbol means that the value of S is **True** when evaluated at the element a ;
- So, the positive integers are a *function*!

Yet, given a function $P : \alpha \rightarrow \text{Prop}$ we prefer to write $\text{setOf } P : \text{Set } \alpha$ to denote the set, rather than $P : \text{Set } \alpha$, to avoid *abusing definitional equality*.

Some examples:

1. How to prove that something belongs to a set?
2. Positive naturals;
3. Even numbers;
4. An abstract set of α given by some $P : \text{Prop}$.

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+++ Sub(sub-sub-sub)sets are not treated as sets-inside-sets.

Given a (old-style) set S , what is a subset T of S ? At least two answers:

1. Another set such that $x \in T \Rightarrow x \in S$.
2. A collection of elements of S .

Now,

1. stresses that T is a honest set satisfying some property;
2. stresses that it is a set whose elements "come from" S .

We take the **first approach**: being a subset is *an implication*

```
variable {T S : Set α}
def (T ⊆ S : Prop) := ∀ a, a ∈ T → a ∈ S
```

Yet this does not answer the question "How can I *construct* a subset of a set"? The key is *upgrading* sets to types: $T : \text{Set } S$ for $S : \text{Set } \alpha$ means $T : \text{Set } \uparrow S = \text{Set } (S : \text{Type}^*)$.

Some examples:

1. Double inclusions;
2. Subsets as sets;
3. This coercion from $\text{Set } \alpha$ to Type^* .

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Operations on Sets

+++ Intersection & Union

Intersection

Given sets $S T : \text{Set } \alpha$ have the

```
def (S ∩ T : Set α) := fun a ↦ a ∈ S ∧ a ∈ T
```

- Often need **extensionality**: equality of sets can be tested on elements;
- related to *functional extensionality*: two functions are equal if and only they have if they take the same values on same arguments;
- not strange: sets *are* functions.

Union

Given sets $S \ T : \text{Set } \alpha$ we have the

```
def (S U T : Set α) := fun a ↦ a ∈ S ∨ a ∈ T
```

And if $S : \text{Set } \alpha$ but $T : \text{Set } \beta$? **ERROR!**

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+++ **Universal & Empty set, complement and difference**

Universal & Empty

- The first (containing all terms of α) is the constant function $\text{True} : \text{Prop}$

```
def (univ : Set α) := fun a ↦ True
```

- The second is the constant function $\text{False} : \text{Prop}$

```
def (∅ : Set α) := fun a ↦ False
```

Bonus: There are infinitely many empty sets!

Complement and Difference

- The complement is defined by the negation of the defining property, denoted S^c .

```
Sc = {a : α | ¬a ∈ S}
```

The superscript ^c can be typed as $\backslash^c$.

- The difference $S \setminus T : \text{Set } \alpha$, corresponds to the property

```
def (S \ T : Set α) = fun a ↦ a ∈ S ∧ a ∉ T
```



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+++ Indexed Intersections & Indexed Unions

- One can allow for fancier indexing sets (that will actually be **types**, *ça va sans dire*): given an index type I and a collection $A : I \rightarrow \text{Set } \alpha$, the union $(\bigcup i, A i) : \text{Set } \alpha$ consists of the union of all the sets $A i$ for $i : I$.
- Similarly, $(\bigcap i, A i) : \text{Set } \alpha$ is the intersection of all the sets $A i$ for $i : I$.
- These symbols can be typed as $\backslash U = \bigcup$ and $\backslash I = \bigcap$.



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Filters

Definition

A filter F on a type α is set in $\text{Set } \alpha$ (*i. e.* a collection of sets in α) such that:

1. The largest set $T = \text{Set.univ}$ is in F ;
2. If $s, t : \text{Set } \alpha$ and $s \subseteq t$, then $s \in F$ implies that $t \in F$ (they are "upwards closed")
3. F is stable by finite intersections.

More precisely, **Filter** is a structure:

```
structure Filter (α : Type*) : Type*
| sets : Set (Set α)
| univ_sets : univ ∈ self.sets
| sets_of_superset : ∀ {x y : Set α}, x ∈ sets → x ⊆ y → y ∈ sets
| inter_sets : ∀ {x y : Set α}, x ∈ sets → y ∈ sets → x ∩ y ∈ sets
```

+++ Some examples of filters

- Given a term $a : \alpha$, the collection of all sets containing a is the **principal** filter (at a): this generalises to any set $S \subseteq \alpha$, being the case $S = \{a\}$. It is denoted $\mathcal{P} S$, typed $\backslash MCP S$.
- The collection of all sets of natural integers (or real numbers, or rational numbers...) that are "large enough" or "small enough" are filters. They are called **atTop** and **atBot**, respectively.
- In a topological space X , the collection of all neighbourhoods (*i. e.* sets containing an open neighbourhood) of a subspace S is a filter, denoted $\mathcal{N} S$; when $S = \{x\}$, we write $\mathcal{N} x$.



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Filters and properties

If $F : \text{Filter } \mathbb{N}$ is a filter on the naturals and $P : \mathbb{N} \rightarrow \text{Prop}$ is a property, we might want to say that

- P holds for every large enough n (e.g. satisfying $n^2 - 5n \geq 0$)
- that we can find arbitrary large n satisfying P (e.g.: being prime).
- We could also say that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $P : \mathbb{R} \rightarrow \text{Prop}$ "for all x sufficiently close to 1" (e.g. being positive, when $f \ 1 > 0$ and f is continuous).

+++ More generally

Given $F : \text{Filter } \alpha$ and $P : \alpha \rightarrow \text{Prop}$, we write

- $\forall^f (a : \alpha) \text{ in } F, P \ a \leftrightarrow \{a : \alpha \mid P \ a\} \in F$
- $\exists^f (a : \alpha) \text{ in } F, P \ a \leftrightarrow \{a : \alpha \mid \neg P \ a\} \notin F$

and we say that P holds **eventually** (resp. **frequently**) with respect to F . The above cases correspond to $\text{atTop} : \text{Filter } \mathbb{N}$ and $\mathcal{N} \ 1 : \text{Filter } \mathbb{R}$.

- Similarly, given $f \ g : \alpha \rightarrow \beta$ and $F : \text{Filter } \alpha$ we write $f =^f[F] \ g$ and we say that f and g are asymptotically equal (along F) if

$$\{a : \alpha \mid f \ a = g \ a\} \in F$$

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Limits

Filters are (among other things) a very convenient way to talk about **convergence**.

Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ and $a, b \in \mathbb{R}$. To say that

$\lim_{x \rightarrow a} f(x) = b$

means

$\forall \varepsilon > 0, \exists \delta > 0 ; \text{such that } ; \|x - a\| < \delta \Rightarrow \|f(x) - b\| < \varepsilon$

or, equivalently,

$\forall \varepsilon > 0, \exists \delta > 0 ; \text{such that } ; f(a - \delta, a + \delta) \subseteq (b - \varepsilon, b + \varepsilon).$

or, equivalently, that

$\forall U_b \in \mathcal{N}_b, \exists V_a \in \mathcal{N}_a ; a \text{ such that } V_a \subseteq f^{-1} U_b.$

Upwards-closeness of filters makes the explicit description of V_a useless: to require $V_a \subseteq f^{-1} U_b$ is the same as

$$\forall U : \text{Set } \mathbb{R}, U \in \mathcal{N}_b \rightarrow f^{-1} U \in \mathcal{N}_a$$

And the statement

$\lim_{x \rightarrow +\infty} f(x) = b$ simply becomes

$$\forall U : \text{Set } \mathbb{R}, U \in \mathcal{N} \quad b \rightarrow f^{-1} \cdot U \in (\text{atTop} : \text{Filter } \mathbb{R})$$

+++ Is this translation really useful?

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ and $a, b, c \in \mathbb{R}$. One theorem is that

$\lim_{x \rightarrow a} f(x) = b \Rightarrow \lim_{y \rightarrow b} g(y) = c \Rightarrow \lim_{x \rightarrow a} (g \circ f)(x) = c$

while

$\lim_{x \rightarrow +\infty} f(x) = b \Rightarrow \lim_{y \rightarrow b} g(y) = c \Rightarrow \lim_{x \rightarrow +\infty} (g \circ f)(x) = c$

is *another* theorem, because $+\infty \notin \mathbb{R}$. And

$\lim_{x \rightarrow a^-} f(x) = -\infty \Rightarrow \lim_{y \rightarrow -\infty} g(y) = c \Rightarrow \lim_{x \rightarrow a^-} (g \circ f)(x) = c$

is a third one. There are (at least) $5^3 = 125$ such theorems.

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Filters as generalised sets

(Recall: elements of $\mathcal{P} \mathbf{s}$ = all sets

containing $\mathbf{s}$.)

• $\mathcal{P} \mathbf{s}$ replaces $\mathbf{s}$, more general filters are "generalised sets" of α .

1. The **order** relation: sets on α are

ordered by inclusion, so $S_1 \leq S_2 \leftrightarrow S_1 \subseteq S_2 \leftrightarrow \forall T, T \supseteq S_2 \rightarrow T \supseteq S_1$. Hence:

$$\text{def le } (F \ G : \text{Filter } \alpha) : F \leq G \leftrightarrow \forall t \in G, t \in F := \text{Iff.rfl}$$

2. Image of a filter through a function $f : \alpha \rightarrow \beta$. This operation is called

Filter.map:

$$\text{theorem mem_map } (t : \text{Set } \beta) (F : \text{Filter } \alpha) : t \in F.\text{map } f \leftrightarrow f^{-1} \cdot t \in F := \text{Iff.rfl}$$

3. With all this, the statement $\lim_{x \rightarrow a} f(x) = L$ becomes

```
def Tendsto (f :  $\alpha \rightarrow \beta$ ) (F : Filter  $\alpha$ ) (G : Filter  $\beta$ ) :=  
  ( $\mathcal{N}$  a).map f  $\leq$  ( $\mathcal{N}$  L)
```

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